Abstract

Abstract I

This paper presents a strongly-typed, decentralized multiagent simulation — an ant-robot colony — as the computational exemplar of the Research Project Template. Each colony member is an `Agent` that owns exactly one real, on-disk SQLite database and one in-process, fault-injectable protocol endpoint; no agent ever touches another agent's storage or network state. The implementation lives under `projects/templates/template_formal/src/template_formal/`; the demo pipeline is orchestrated by `scripts/02_run_analysis.py`.

Abstract II

The paper's central claim is methodological, not a typing-features showcase: static typing's honest value in Python is edit-time/CI-time error prevention on structurally representable invariants, and nothing more. Nominal identifiers (`AgentId`, `MessageId`, `TxnId` as distinct `NewType` wrappers), a tagged-union `Result[T, E]` ADT with `match`-exhaustiveness, and a session-typed protocol state machine (`IdleSession` → `HandshakingSession` → `EstablishedSession` → `ClosedSession`) each make an illegal program a **type error**, verified by a real `mypy --strict` subprocess run against six known-bad negative-control fixtures plus three known-good positive-control fixtures (`tests/mypy_fixtures/`). Where the type system cannot help — reusing a consumed transaction handle, reusing a consumed protocol-phase instance, or receiving malformed bytes off an untyped network boundary — the implementation runtime-guards instead, and the manuscript says so explicitly rather than eliding the distinction.

Abstract III

We also frame, without over-claiming, two additional lenses: the per-agent storage schema as a functor $\mathrm{Schema} \to \mathbf{Set}$ in the sense of @fong2018seven, and each agent's per-tick decision as an approximate minimizer of a closed-form expected-free-energy quantity in the spirit of @friston2005theory, bridged to collective organization via the Memory Evolutive Systems framework of [@ehresmann2007memory]. Both framings are declared as design lenses, not machine-checked mathematical results — the paper is explicit about which of its claims are proofs and which are analogies.

Abstract IV

Contributions are architectural, epistemic, and empirical. Architecturally: a zero-mock test suite (`tests/`) covering ADT exhaustiveness, affine-handle reuse, session-type phase transitions, seeded fault injection over a real in-process bus, and a three-agent colony integration test exhibiting a real stigmergic positive-feedback mechanism (deliberately not overclaimed as "emergence" — see @sec:results-discussion). Epistemically: an explicit "What mypy –strict proves vs. what is a runtime discipline" section (@sec:honesty-line) that pins every strong claim to the ISC (Ideal-State Criterion) number of its paired negative-control test, so the claim-to-evidence mapping is auditable rather than asserted. Empirically: eleven pre-registered analyses grouped across three experiment families, falsifiable experiments (@sec:results-discussion) — a decay-rate sweep revealing a real, non-monotonic threshold effect (near-zero convergence below decay ≈ 0.35, a 100% plateau at moderate decay, and a measurable decline at total evaporation); a random-choice null-model comparison showing the real mechanism's Wilson-bounded convergence rate (93.3%) does not overlap a chance baseline's (0.67%); and a